

A Short Guide to Lemma D.2

Følner averaging and exact covariance in the Gaussian model

Abstract

Lemma D.2 converts an optimization averaged over larger and larger boxes of displacements into an optimization at the origin over exactly covariant maps. This is a noncompact-group version of symmetrizing a channel. The main technical point is that averaged channels need not have a trace-preserving limit in infinite dimension; the limit can be subnormalized because probability may escape to arbitrarily high energy. This note supplies the background and expands the compactness argument that is only referenced in the paper's short proof.

1 Why the reduction is useful

The Gaussian cloning problem asks for a channel that performs well on every displaced input

$$\Phi_z = D(z)\Phi_0 D(z)^*, \quad z \in \mathbb{C}^{r_d},$$

against the correspondingly displaced target

$$\Psi_z = D(\sqrt{\gamma}z)\Psi_0 D(\sqrt{\gamma}z)^*.$$

The flat-prior criterion averages the root fidelity over a large cube of z 's. A general channel may behave differently at different z . Lemma D.2 shows that, in the large-cube limit, it is enough for an upper bound to consider maps satisfying the exact covariance relation

$$\Gamma(D(z)X D(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^*.$$

For such a map, performance is the same at every displacement, so it can be evaluated at $z = 0$.

2 Background

2.1 Root fidelity and symmetry averaging

For positive trace-class operators R, S , possibly subnormalized, the root fidelity is

$$\mathsf{B}(R, S) = \mathrm{Tr} \sqrt{R^{1/2} S R^{1/2}}.$$

The two properties used here are

$$\begin{aligned} \mathsf{B}(URU^*, USU^*) &= \mathsf{B}(R, S), \\ \mathsf{B}\left(\int R_z \, \mathrm{d}\mu(z), \int S_z \, \mathrm{d}\mu(z)\right) &\geq \int \mathsf{B}(R_z, S_z) \, \mathrm{d}\mu(z), \end{aligned}$$

for a unitary U and a probability measure μ . The second line is joint concavity. It says that averaging conjugated copies of a channel cannot decrease its average fidelity against the correspondingly conjugated targets.

2.2 Følner cubes

Identify $\mathbb{C}^{r_d}$ with the additive group $\mathbb{R}^{2r_d}$, and let

$$Z_L = \{z : |\Re z_\ell| \leq L, |\Im z_\ell| \leq L \text{ for every } \ell\}.$$

For every fixed $v \in \mathbb{C}^{r_d}$,

$$\frac{|(Z_L + v) \triangle Z_L|}{|Z_L|} \longrightarrow 0. \quad (1)$$

Here $\triangle$ denotes symmetric difference. This is the Følner property: shifting a large cube changes only a boundary layer, whose relative volume tends to zero.

For a compact group one could average over the whole group using normalized Haar measure and obtain an exactly covariant channel immediately. Phase space is noncompact and has no invariant probability measure. Følner averaging replaces that unavailable global average by averages over expanding cubes, followed by a compactness limit.

2.3 Why weak-* compactness is needed

The multimode Fock space $\mathcal{F}_{\text{orb}}$ is infinite-dimensional. Its trace class is the dual of the compact operators:

$$\mathcal{T}_1(\mathcal{F}_{\text{orb}}) = \mathcal{K}(\mathcal{F}_{\text{orb}})^*.$$

Closed norm-bounded sets of trace-class operators are compact in the weak-* topology coming from tests against compact operators, though not in trace norm. In this topology trace can be lost. For instance,

$$|n\rangle\langle n| \xrightarrow{w^*} 0 \quad \text{on } \mathcal{K}(\mathcal{F}_{\text{orb}}),$$

although every term has trace one. This describes mass escaping to arbitrarily high photon number.

To remember that the approximating outputs are normalized, one works first with the unitization

$$\tilde{\mathcal{K}} = \mathcal{K}(\mathcal{F}_{\text{orb}}) \oplus \mathbb{C}I.$$

A weak-* limit is a state on $\tilde{\mathcal{K}}$. Its restriction to $\mathcal{K}(\mathcal{F}_{\text{orb}})$ is represented by a positive trace-class operator of trace at most one; the missing trace is the component at the quotient character, or “infinity.” A normalized trace-class target has no component there, so the escaped mass has zero fidelity with it.

3 Statement of Lemma D.2

Define the input and output actions

$$\alpha_z^{\text{in}}(X) = D(z)XD(z)^*, \quad (2)$$

$$\alpha_z^{\text{out}}(X) = D(\sqrt{\gamma}z)XD(\sqrt{\gamma}z)^*. \quad (3)$$

Then $\Phi_z = \alpha_z^{\text{in}}(\Phi_0)$ and $\Psi_z = \alpha_z^{\text{out}}(\Psi_0)$.

Lemma D.2 (rephrased). *For the centered cubes Z_L ,*

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) dz \leq \sup_{\Gamma} B(\Gamma(\Phi_0), \Psi_0). \quad (4)$$

The left supremum is over quantum channels. The right supremum is over completely positive trace-nonincreasing maps satisfying

$$\Gamma \circ \alpha_z^{\text{in}} = \alpha_z^{\text{out}} \circ \Gamma \quad (z \in \mathbb{C}^{rd}).$$

The limiting map Γ may have subnormalized output.

The inequality is in the direction needed for a converse: every sequence of nearly optimal channels on expanding cubes produces an exactly covariant competitor on the right.

4 Proof walkthrough

4.1 Step 1: average a channel over a finite cube

For a channel Λ , define

$$\bar{\Lambda}_L = \frac{1}{|Z_L|} \int_{Z_L} \alpha_{-z}^{\text{out}} \circ \Lambda \circ \alpha_z^{\text{in}} \, dz. \quad (5)$$

Every integrand is a channel, so $\bar{\Lambda}_L$ is a channel. Joint concavity and unitary invariance of root fidelity give

$$\begin{aligned} B(\bar{\Lambda}_L(\Phi_0), \Psi_0) &\geq \frac{1}{|Z_L|} \int_{Z_L} B(\alpha_{-z}^{\text{out}} \Lambda(\Phi_z), \Psi_0) \, dz \\ &= \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) \, dz. \end{aligned} \quad (6)$$

Thus the original average payoff is bounded by the origin payoff of the averaged channel.

4.2 Step 2: extract a weak-* limit of averaged channels

Choose $L_j \rightarrow \infty$ and channels Λ_j whose payoffs approach the left side of (4). Let $\bar{\Lambda}_j$ denote their averages. The adjoints restricted to the unitized compact operators are unital completely positive contractions

$$\bar{\Lambda}_j^* : \tilde{\mathcal{K}} \longrightarrow \mathcal{B}(\mathcal{F}_{\text{orb}}) = \mathcal{T}_1(\mathcal{F}_{\text{orb}})^*.$$

The algebra $\tilde{\mathcal{K}}$ is separable. On bounded subsets of $\mathcal{B}(\mathcal{F}_{\text{orb}})$, the weak-* topology is metrizable because the predual $\mathcal{T}_1(\mathcal{F}_{\text{orb}})$ is separable. A diagonal argument on a countable dense operator system therefore produces a subsequence and a pointwise weak-* limit

$$\hat{\Gamma}^* : \tilde{\mathcal{K}} \longrightarrow \mathcal{B}(\mathcal{F}_{\text{orb}}).$$

Positivity at every matrix level is weak-* closed, so $\hat{\Gamma}^*$ is completely positive; convergence at the identity shows that it is unital.

Restrict $\hat{\Gamma}^*$ to $\mathcal{K}(\mathcal{F}_{\text{orb}})$ and define $\Gamma : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \rightarrow \mathcal{T}_1(\mathcal{F}_{\text{orb}})$ by

$$\text{Tr}[\Gamma(X)A] = \text{Tr}[X\hat{\Gamma}^*(A)], \quad A \in \mathcal{K}(\mathcal{F}_{\text{orb}}). \quad (7)$$

This gives a completely positive map. Let (e_n) be increasing finite-rank projections converging strongly to I . For $X \geq 0$,

$$\begin{aligned} \text{Tr} \Gamma(X) &= \sup_n \text{Tr}[\Gamma(X)e_n] \\ &= \sup_n \text{Tr}[X\hat{\Gamma}^*(e_n)] \leq \text{Tr}[X\hat{\Gamma}^*(I)] = \text{Tr} X. \end{aligned}$$

Hence Γ is trace nonincreasing. Equality need not hold because the restriction from $\tilde{\mathcal{K}}$ to $\mathcal{K}(\mathcal{F}_{\text{orb}})$ discards possible mass at infinity.

4.3 Step 3: the Følner property gives exact covariance

For fixed $v \in \mathbb{C}^{r_d}$, changing variables in (5) gives

$$\left\| \bar{\Lambda}_L(\alpha_v^{\text{in}} X) - \alpha_v^{\text{out}} \bar{\Lambda}_L(X) \right\|_1 \leq \frac{|(Z_L + v) \triangle Z_L|}{|Z_L|} \|X\|_1. \quad (8)$$

The integrands are trace-norm contractions, and only the two boundary pieces left by shifting the cube contribute. By (1), the right side tends to zero. Pairing (8) with compact operators and passing to the extracted weak-* limit yields

$$\Gamma(\alpha_v^{\text{in}} X) = \alpha_v^{\text{out}} \Gamma(X).$$

This holds for every fixed v . One may first prove it on a countable dense set of v 's used in the diagonal extraction; strong continuity of the Weyl operators, which gives trace-norm continuity of their action on trace class, then extends it to all v .

4.4 Step 4: pass the fidelity payoff to the limit

Put $\rho_j = \bar{\Lambda}_j(\Phi_0)$. Viewed as states on $\tilde{\mathcal{K}}$, these converge weak-* to a state $\hat{\rho}$. Its restriction to $\mathcal{K}(\mathcal{F}_{\text{orb}})$ is the trace-class operator

$$\rho = \Gamma(\Phi_0),$$

which may have trace less than one.

For positive functionals on a unital C^* -algebra, the Alberti–Uhlmann variational formula is

$$B(\phi, \psi) = \inf_{A > 0} \inf_{\text{invertible}} \frac{1}{2} (\phi(A) + \psi(A^{-1})). \quad (9)$$

For fixed ψ , the right side is an infimum of weak-* continuous affine functions of ϕ , so it is weak-* upper semicontinuous. Therefore

$$\limsup_j B(\rho_j, \Psi_0) \leq B_{\tilde{\mathcal{K}}}(\hat{\rho}, \hat{\Psi}_0).$$

The target Ψ_0 is normalized and supported on the original compact operator algebra. Its extension has zero component at infinity. The central-summand, or mass-at-infinity, identity therefore gives

$$B_{\tilde{\mathcal{K}}}(\hat{\rho}, \hat{\Psi}_0) = B(\Gamma(\Phi_0), \Psi_0).$$

Combining this inequality with (6), and taking the supremum over all possible covariant limits Γ , proves (4).

5 What subnormalization means

Subnormalization is not an extra physical resource and does not enlarge the original class of channels before taking the limit. Every $\bar{\Lambda}_j$ is trace preserving. The trace defect appears only because compact observables cannot detect output weight that moves to unbounded photon number. Such weight has vanishing overlap with the fixed trace-class target, so retaining it would not improve the limiting fidelity.

The elementary sequence $|j\rangle\langle j|$ illustrates the point. Its weak-* limit on compact operators is zero, while on the unitization the missing unit trace survives entirely at the quotient character. Against a fixed normalized target in the original Fock space, that quotient component is orthogonal and contributes no fidelity.

6 Dependencies and role in the paper

Lemma D.2 is the purely quantum specialization of Lemma C.2. Relative to that hybrid result, the classical coordinate is omitted and the translation group is only $\mathbb{C}^{r_d}$. Its proof uses:

1. joint concavity and unitary invariance of root fidelity;
2. the Følner property of Euclidean cubes;
3. weak-* compactness for completely positive maps on the unitized compact-operator algebra;
4. upper semicontinuity of transition amplitude and the mass-at-infinity identity explained in Lemma C.1.

It does not use Lemma D.1. Lemma D.1 enters the separate stochastic-order chain beginning with Lemma D.3.

In the proof of Theorem 3.4, the product quantum-limited amplifier supplies the lower bound. For the converse, Lemma D.2 reduces the large flat-prior average to

$$B(\Gamma(\Phi_0), \Psi_0)$$

for an exactly displacement-covariant trace-nonincreasing map. Phase twirling then makes the map phase covariant without lowering this origin fidelity, and Lemma D.5 supplies the explicit multimode upper bound. Thus D.2 is the symmetry-and-compactness step that makes the later photon-number argument applicable.

7 Minimal prerequisite checklist

The conceptual core requires only group averaging, concavity of fidelity, and the fact that large cubes have negligible relative boundary. The functional-analytic part additionally uses trace-class/compact-operator duality, weak-* compactness, unitization, and upper semicontinuity of the Alberti–Uhlmann transition amplitude. Its practical message is simple: an asymptotically translation-invariant average may be replaced, for an upper bound, by one exactly covariant map evaluated at the origin.